Practical No.5

Aim: Design and develop JSP application to demonstrate JSP Scripting elements,JSP Directives , JSP Implicit Objects 4. JSP Action tags.

Theory: JSP is a server-side technology used to create dynamic web pages using HTML and Java. JSP allows embedding Java code inside HTML pages to generate dynamic content.

- **JSP Scripting Elements:** JSP scripting elements allow Java code to be inserted into a JSP page.Types of JSP Scripting Elements:

a) Scriptlet : Used to write Java code. **Syntax:**
<% Java code %>
Example:
<%
int a = 10;
int b = 20;
out.println("Sum = " + (a + b));
%>
	b) Expression : Used to display output directly on the browser.
	Syntax:
	<%= expression %>
	Example:
	<%= "Welcome to JSP Programming"
	%>
		c)Declaration : Used to declare variables and methods.
		Syntax:
		<%!declaration %>
		Example:
		<%!
		int square(int n) {
		return n * n;
		}
		%>

- **JSP Directives :**Directives provide instructions to the JSP container about how the JSP page should be processed. Types of JSP Directives:
  a) Page Directive: Defines page-level attributes.
  Syntax: <%@ page language="java" contentType="text/html" %>
  Example: <%@ page import="java.util.Date" %>
  b) Include Directive: Includes another file at translation time.
  Syntax: <%@ include file="header.jsp" %>
  c) Taglib Directive: Used to define custom tag libraries.
  Syntax:<%@ taglib uri="uri" prefix="p" %>
- **JSP Implicit Objects :** Implicit objects are predefined objects created by the JSP container and available without declaration.Common JSP Implicit Objects:

Object	Description
request	Holds client request data
response	Sends response to client
session	Stores user session data
application	Application-wide data
out	Writes output to browser
pageContext	Page-related info

- **JSP Action Tags:** Action tags are used to perform specific actions at request time.

  a) **jsp:include**: Includes another JSP at runtime.

  Syntax: <jsp:include page="footer.jsp" />

  b) **jsp:forward**: Forwards request to another resource.

  Syntax:<jsp:forward page="welcome.jsp" />

  c) **jsp:useBean**:Creates or accesses a JavaBean.

  Syntax:<jsp:useBean id="obj" class="mypack.MyBean" />

  d) **jsp:setProperty**:Sets property value of a JavaBean.

  Syntax:<jsp:setProperty name="obj" property="*" />

  e) **jsp:getProperty**:Gets property value from a JavaBean.

  Syntax:<jsp:getProperty name="obj" property="name" />

Conclusion:

Problem Statement:

1. Design a JSP-based Student Information Display System

* **Problem Description:** Develop a web application using **Java Server Pages (JSP)** to collect and display student details such as name, roll number, course, and marks.

* **Requirements:**

- Use **HTML form** to collect student details.
- Use **Scriptlet Tag (<% %>)** to:
  - Retrieve form data using implicit objects.
  - Perform total and average calculation.
- Use **Declaration Tag (<%! %>)** to:
  - Declare a method to calculate grade based on average.
- Use **Expression Tag (<%= %>)** to:
  - Display student details, total, average, and grade.
- Use **Implicit Objects**:
  - request → to get form values.
  - out → to print output.
  - session → to store student name.
  - application → to count number of students visited the page.

***Expected Features:**

1. Display formatted result page.
2. Show grade (A/B/C/Fail).
3. Maintain visit counter using application object.

****

OR

2: Simple Online Shopping Bill Generator

* Problem Description:Create a JSP page that allows users to select products and quantities, then generates a bill.

* Requirements:

- Use **Scriptlet Tag** to:
  - Calculate total bill amount.
  - Apply discount if total > ₹5000.
- Use **Declaration Tag** to:
  - Declare a tax calculation method.
- Use **Expression Tag** to:
  - Display item names, quantities, total amount, tax, and final amount.

- Use **Implicit Objects**:
  - request → to retrieve product and quantity.
  - response → to redirect to thank-you page.
  - session → to store customer name.
  - application → to maintain total customers visited.

* Expected Output:

2) Display detailed bill with GST.
3) Show discount if applicable.
4) Display visit counter.

3:Demonstration of JSP Directives:1.Page 2.Include

directives.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1" import="java.util.Date"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
        <h1>Demonstration of @page directive</h1>
        <h2> Current Date and Time</h2>
        <font color="blue" size=18>
        <%= new Date() %>
        </font>
        <%@ include file="include.jsp"  %>
</body>
</html>
```

include.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
```

```
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<h1>Demonstration of @include directive</h1>
</body>
</html>
```

4:Demonstration of JSP Action Tags:<jsp:include>

include_action.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<h1>Action Tags in JSP</h1>
<jsp:include page="action_tag1.jsp"></jsp:include>
</body>
</html>
```

action_tag1.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<h1><FONT color="blue">Demonstration of jsp include action
tag</FONT></h1>
</body>
</html>
```

5:Demonstration of JSP Action Tags:<jsp:forward>

forward.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
```

```
<form action="forward.jsp" method="post">
  Username: <input type="text" name="user">
  <input type="submit" value="Login">
</form>
</body>
</html>
```

forward.jsp

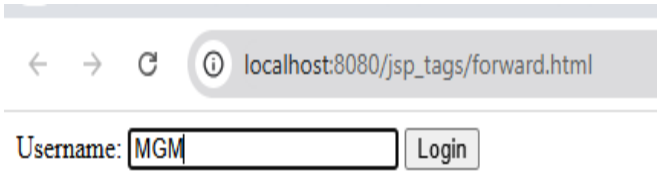
```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<%
String user = request.getParameter("user");
if(user == null || user.equals("")) {
%>
  <jsp:forward page="error.jsp" />
<%
} else {
%>
  <jsp:forward page="success.jsp">
   <jsp:param name="username" value="<%= user %>" />
   </jsp:forward>
<%
}
%>
</body>
</html>
```

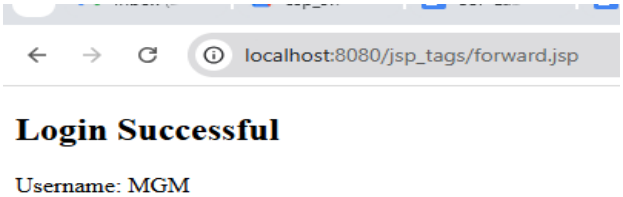
success.jsp

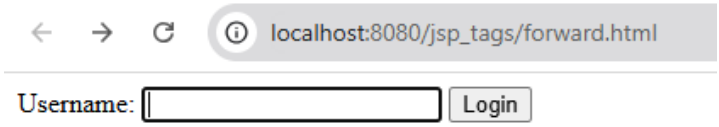
```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<h2>Login Successful</h2>
Username: <%= request.getParameter("username") %><br>
</body>
</html>
```

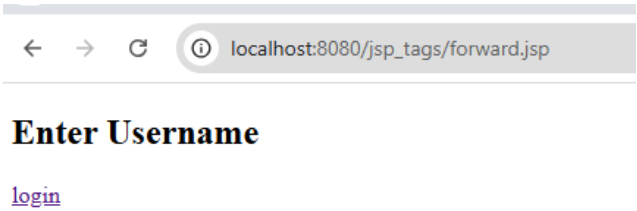
error.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<h2>Enter Username</h2>
<a href="forward.html">login</a>
</body>
</html>
```









6:Demonstration of JSP Action Tags:<jsp:usebean><jsp:setproperty><jsp:getproperty>

student.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<form action="Usebean.jsp" method="post">
  Student ID:
  <input type="text" name="id"><br><br>
  Student Name:
  <input type="text" name="name"><br><br>
  <input type="submit" value="Submit">
</form>
</body>
</html>
```

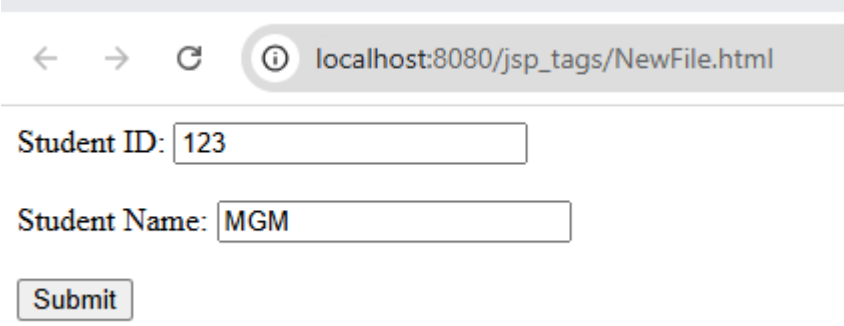
UsebeanDemo.java

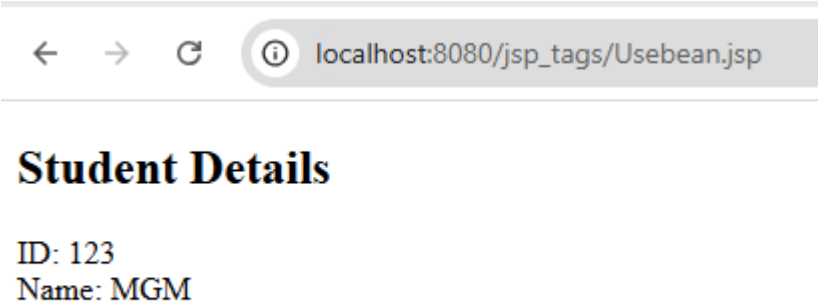
```
package p1;
public class UsebeanDemo {
        private int id;
          private String name;
          public UsebeanDemo() {}
          public int getId() {
             return id;
          }
          public void setId(int id) {
             this.id = id;
          }
          public String getName() {
             return name;
          }
          public void setName(String name) {
             this.name = name;
          }
}
```

Usebean.jsp

```
<%@ page language="java" contentType="text/html; charset=ISO-8859-1"
  pageEncoding="ISO-8859-1"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="ISO-8859-1">
<title>Insert title here</title>
</head>
<body>
<jsp:useBean id="s"   class="p1.UsebeanDemo"    scope="request" />
<jsp:setProperty   name="s"   property="*"  />
<h2>Student Details</h2>
ID: <jsp:getProperty   name="s"  property="id" /><br>
Name: <jsp:getProperty   name="s"  property="name" />
```

```
</body>
</html>
```


localhost:8080/jsp_tags/NewFile.html
Student ID: 123
Student Name: MGM
Submit


localhost:8080/jsp_tags/Usebean.jsp
Student Details
ID: 123
Name: MGM